Behavioral and Genetic Models of Substance Abuse, ADHD, and Amphetamine Use

Prescription amphetamine's usage and abuse has skyrocketed in recent years, following the rise in ADHD diagnoses and subsequent circulation of these psychostimulants (Varga, 2012, p. 295-296). At the correct dosages, these stimulants, such as dextroamphetamine (Dexedrine), methylphenidate (Ritalin), and amphetamine-dextroamphetamine (Adderall), treat symptoms of inattention and hyperactivity in those with ADHD, but they can be used recreationally to induce intoxication with symptoms such as euphoria if used other than how they were prescribed. Amongst young student populations, ranging from middle school to high school, this kind of abuse has become increasingly commonplace, stemming from desires for increased academic success due to its perceived benefits for concentration or, alternatively, desires for intoxication (Varga, 2012, p. 294-295). This increasing prevalence of amphetamine abuse is problematic due to its high potential for addiction. Frequent usage also quickly increases one's tolerance, necessitating constant increases in dosage in order to achieve the same psychological effects, resulting in usage of higher and more dangerous dosages. Prolonged high dose amphetamine use can induce psychosis, due to the paranoia and anxiety inducing effects of this dosage (O'Malley & O'Malley, 2022). Generally, chronic use has been linked with neurological damage, behavioral changes, and decreased cognitive functioning; further, chronic usage, specifically in adolescents, has shown profound neurological changes, namely with respect to the still-developing prefrontal cortex (Wikipedia, 2023). The aforementioned increasing usage of amphetamines and the high risks associated therewith motivate this current review of contemporary research with respect to behavioral and genetic studies of amphetamine use and abuse.

Further motivating this literature review is the large correlation between those who have the highest access to prescription amphetamines, ADHD patients, and substance use disorders (SUDs). Studies have found a high degree of comorbidity of ADHD and SUDs, where up to 20% of adults with a SUD are estimated to have ADHD (Schellekens, 2010). There is also some evidence that compared to those with only SUD, those with both ADHD and SUD develop dependence quicker, start drug use earlier, and have increased relapse rates (Schellekens, 2010). Together these data indicate those with ADHD are at further risk for substance abuse and may have worse outcomes; however, the studies directly studying ties to ADHD and amphetamine use are limited. Therefore, we aim to identify gaps in knowledge in order to call for future research on this important issue.

Beginning our foray into the realm of study methods we arrive at the two main perspectives taken when studying patterns of amphetamine addiction: Genetic and Behavioral. Nestler summarizes the inputs related to systems of addiction as "Genetic and environmental factors, by establishing all aspects of normal synaptic structure and function, determine an individual's inherent sensitivity to initial drug exposure. These factors also establish how individual nerve cells, and the circuits in which they operate, adapt over time to chronic drug exposure, which in turn determines the development of addiction." (Nestler, 2000). This perspective is of great importance as, through it, we can highlight the major obstacles to understanding and identifying genes associated with addiction.

The single most important genetic obstacle of note, under which most other obstacles fall, is that addiction is a highly complex trait. Due to the nature of polygenic traits like addiction, single genes usually produce very small changes in expression of the trait which provides a great challenge in experimentally detecting these changes. Epigenetic effects are also quite important to the discussion of addiction as "environmental factors such as stress can interact with an animal's genotype to determine the ultimate response to a drug of abuse." (Nestler, 2000). Continuing on the discussion of environmental and epigenetic factors, it is also important to note another important genetic roadblock to studying addiction: lineage. Because of how heterogeneous and polygenic in nature addiction is, these genetic variants can present themselves differently in different generations and even different members of the same generation of a family with known genetic linkages to addiction. This provides a very difficult landscape to navigate with regard to understanding what genes play a role in addiction and addictive tendencies as even within the same family with the same genetic modifications these genes can be of varying levels of importance to the overall susceptibility of an individual to addiction.

Moving now in the direction of methods of experimental measurement with regard to the genetics of addiction we can see a wide range of techniques and tactics for determining the genetic inputs to and results of addiction. There are four main methods of genetic study of addiction as outlined in Nestler's conglomeration and summary of common practice in *Nature Genetics*. All of these methods utilize mice as the model organism. The first common practice is examining the results of Constitutive Mutations such as knockouts and analysis of overexpression. By examining the results of either knocking out a piece of genetic material thought to have linkages to addiction or influencing/examining overexpression of these suspected contributor gene(s), clinical experimenters can garner insight into the effects of each chunk of genetic material they are testing on addiction and addictive tendencies. The second common genetic technique is quite similar in nature to the first. The second technique is Inducible and Cell-type Specific Mutations. This technique utilizes the same frames of examination (knockouts and overexpression) but induces them in a way that can be controlled.

As a result, the duration of the mutation being present can be controlled as well as the type of cell in which the mutation is expressed. This provides quite similar data than the first technique discussed but allows researchers to examine the effects on a much smaller and more specific scale as they can refine their hypotheses surrounding influences on addiction. The third technique is most likely the most common method of genetic study: Viral-Mediated Gene Transfer. By utilizing viruses, researchers can implant certain mutations in the genetic code of the subject as a result of the virus' method of RNA transmittance as it incorporates its genetic code into that of the targeted cell. This technique provides a circumstance with very "Highly circumscribed spatial and temporal resolution of mutation" (Nestler, 2000) allowing researchers to delve much deeper into the mechanics of induced addiction and examine very specific cellular relationships underlying addictive behaviors/tendencies. The fourth and final common method of study is Antisense Oligonucleotides. Utilizing these oligonucleotides, researchers can modify cellular RNA sequences interfering with normal function and examining the results of this disruption. Because oligonucleotides function by binding to their complementary RNA sequence, they can be made highly specialized to target very specific sequences that are associated with proteins with known or expected involvement in addiction to observe the results of modifying those proteins or getting rid of them altogether. Antisense Oligonucleotides can affect Gene Expression Regulation, Splicing Modulation, and cause RNA Degradation, all of which can be used to study the mechanics of addiction and provide phenomenal spatial and temporal resolution of the intended mutations as well as rapid development times in comparison to mutant model organisms (usually mice).

To address our final pathway of study, we look toward behavioral methods. A large subset of these techniques were presented and analyzed in Phillips et. al's 2008 literature summary “Behavioral genetic contributions to the study of addiction-related amphetamine effects” but we will choose to delve into the four most common methods; keeping in line with previous discussion. Behavioral methods of study revolve around observation post-administration to track trends and discover how addiction modulates behavior. The first major method of study is Acute Locomotor Stimulation. This method of study uses mice as model organisms as their locomotor stimulation as a result of amphetamine administration has been suggested to model the euphoria experienced by humans under similar drug-induced circumstances (Phillips et. al., 2008). By modulating experimental conditions (drug concentration, administration patterns, administration location, etc.), experimenters can model euphoric reactions in humans using the locomotor stimulation in model organisms. The second mode of study is referred to as Stereotypy. Stereotypy is the name given to repetitive movements or actions associated with addiction-induced behaviors. Stereotypy is studied in two different ways. The first of these is with high-dose amphetamine treatment to induce acute Stereotypy which are behaviors normally associated with the drug itself and initial addiction symptoms. The second is with repeated treatment to attempt to induce Stereotypy normally associated with heavy addiction or chronic addictive tendencies. Our third method of study is Locomotor Sensitization. Locomotor Sensitization refers to the process of creating enhanced sensitivity to motoric and stereotypic effects of amphetamines through daily or less-frequent repeated administration (Phillips et. al., 2008). Phillips also noted that the 1997 Bartlett et. al study “...hypothesized that the neuroadaptations underlying behavioral sensitization contribute to the paranoia and psychosis characteristic of psychostimulant abuse.” (Phillips et. al., 2008). The fourth and final method to discuss is studies of Neurotoxicity. Neurotoxicity refers to brain alterations associated with amphetamine abuse that can be detected using imaging techniques.

As a result of detection of recent amphetamine use being linked to changes in dopamine and glutamate transmission, mitochondrial disruption, and amphetamine-induced hyperthermia (Phillips et. al., 2008), study of these particular symptoms and their effects on behavior and the physiological state of the subject can shed light onto the mechanisms behind amphetamine-induced behavioral modulation and other physical repercussions of long term amphetamine abuse.

According to Segal. et al. (1974) we have a rat model of amphetamine use related behavioral consequences. To understand these behavioral consequences, the researchers measured the level of activities before and after the administration of the substance of interest, compared with the baseline, then drew conclusions from the difference. These experiments involved adult rats that were stabilized under the desired experimental condition, being injected with either saline or different dosages (0.5, 1.0 , 2.5, 5.0, and 7.5 mg/kg) of d-amphetamine for 36 days, then continuously monitoring their behaviors. After 36 days, the 1.0 and 7.5 mg/kg dosage rats were injected with saline for an additional 7 days, then on Day 44 they were retested with amphetamine.

The behaviors for the rats that were injected with saline did not differ for the before and after time points. However, the behaviors of rats that were injected with d-amphetamine changed. The increase in stereotypy competed with the expression of locomotor activity in the following 3 to 4 hours after the injection, where for the dosage of 2.5 mg/kg, for the hour following the initial 12-minute period of hyperactivity had significantly reduced locomotion compared to the stereotypy. In addition, the lower-dose rats had a progressive increase in locomotion throughout the 36 days of the experiment, while the higher-dose (5.0 and 7.5 mg/kg) rats had a progressive augmentation of locomotion during the after-phase.

The saline injections from Day 37 to Day 43 showed that the rats produced a response that was similar to that of baseline levels. However, the d-amphetamine injection on Day 44 marked a significant increase in locomotion that was greater than the initial injection on Day 1, indicating that "...augmentation in response to repeated administration of amphetamine persists for at least eight days after discontinuing the chronic regimen" (Segal et al., 1974) for the 1.0 mg/kg dose, and similar for the after-phase augmentation of locomotor activity for the 7.5 mg/kg dose.

The behavioral consequence of administering amphetamine includes the dark phase, which refers to the period of decreased activities for up to 18 hours after the initial hyperactivity from administering high-dosage of amphetamine or repeated low-dosage. This reduction to the subject's locomotion is not secondary to the initial hyperactivity produced by amphetamine, and this progressive behavior change persists in an altered state even if the post-injection activity returns to pre-drug usage levels after the subject stops the chronic amphetamine usage. In conclusion, the usage of amphetamine, high-dosage or repeated low-dosage, will cause augmentation in stereotypy and locomotion in humans similar to what the researchers observed in rats in these experiments. Also, chronic administration of amphetamine can induce a behavioral pattern similar to that of paranoid schizophrenia, and if the dosage is high, it can gradually develop amphetamine-induced psychosis (Segal et al. 1974).

Understanding addiction is multidimensional, studied from behavioral, social, and biological perspectives. The biological or genetic path of enquiry studies genetic factors and differences that give more information about the susceptibility towards addiction. To deeply understand addiction and to work in a way to mediate the process of addiction, model organisms are used in scientific research to comprehend the complexity of the genetic basis of addiction and effectively apply it to the benefit of humans.

Behavioral genetics is a significant field when it comes to studying genetics using model organisms. Scientists have created knockout/altered genes in different animals. All these animals with altered levels of a gene under study are subject to behavioral tests to note the effects of the abused drug. The behavioral responses are measured in the form of increased locomotor activity, which has a direct relation to the mesolimbic dopamine system, a part of the brain that is associated with reward and also addiction. To better understand the role of the mesolimbic dopamine system, classically and operantly conditioned animals were behaviorally studied to determine the relation between drug exposure and place preference, and more complexly- using self-administration, intracranial self stimulation and conditional reinforcement paradigms (Nestler, 2000).

The persistence of behavioral abnormalities in addiction points towards a role for drug-induced changes in gene expression. ΔFosB, a stable transcription factor accumulating in the nucleus accumbens after chronic drug exposure, acts as a sustained molecular 'switch' contributing to addiction. On the other hand, CREB, another transcription factor, plays a dual role- its overexpression in the nucleus accumbens counters drug reward but promotes physical dependence. Identifying target genes for ΔFosB and CREB involves strategies like examining candidate genes, leading to discoveries like the AMPA glutamate receptor subunit GluR2 mediating ΔFosB action and the opioid peptide dynorphin as a target for CREB, partially mediating CREB-induced repression of drug reward (Nestler, 2000).

To narrow down specifically to amphetamine and amphetamine-like drug addiction is to look at literature that studies and analyzes inbred stains, single gene knockouts, transgenics, and QTL (Quantitative Trait Locus) mapping populations.

Quantitative Trait Loci (QTL) mapping, initiated around 15 years ago for studying addiction-related traits in mice, involves identifying chromosomal regions influencing complex traits controlled by genetic and environmental factors. After initial mapping, finer mapping narrows down regions containing genes contributing to trait variation. Functional polymorphisms related to behavioral differences may be identified via sequence analysis, and gene expression analyses help find genes differentially expressed on the basis of behavioral traits. However, existing QTL mapping data for amphetamine-related traits are less and provisional.

The use of Recombinant Inbred (RI) mouse strains has played a crucial role in mapping QTL linked with amphetamine-related traits. Initially designed for major gene mapping, technological advancements have allowed for the exploration of minor gene loci influencing traits mediated by multiple genes. Early studies on hyperthermic response to d-amphetamine in BXD RI mice showed a significant QTL on mouse chromosome 1, indicative of a potential major gene effect. However, subsequent investigations with a larger strain set and different doses suggested dose-specific or false positive influences, bringing in the need for finer mapping for confirmation and gene identification. Other amphetamine-related traits, including stereotyped climbing and acute locomotor activation, have also undergone QTL mapping using RI strains, revealing provisional QTL that require further validation and gene identification. Despite advancements, no definitive quantitative trait genes (QTG) have been identified, highlighting the need for continued research and refinement in understanding the genetic basis of amphetamine-related behaviors in mice (Phillips et al., 2008).

The study by Vilar-Ribó and others (2020) investigated the genetic basis of substance use disorders (SUD) in individuals with ADHD. The researchers employed polygenic risk scores (PRS) from pre-existing GWAS-MA summary statistics for smoking initiation, alcohol dependence, lifetime cannabis use, cocaine dependence, etc. These PRS were tested for association with SUD-related phenotypes in an in-house ADHD cohort using logistic regression and adjusted for various factors.

The results showed significant associations between PRS for lifetime cannabis use, alcohol dependence, and smoking initiation with corresponding SUD-related phenotypes in the ADHD cohort. There was also a positive association between SUD traits and ADHD, found through genetic correlation studies, which suggests a shared genetic load. Mendelian randomization (MR) analyses were conducted to test causality between ADHD and SUD-related phenotypes bidirectionally. The findings supported a causal effect of the genetic liability to ADHD on smoking initiation, age of smoking initiation, and cigarettes per day (Vilar-Ribó et. al., 2020).

The study's results provide insights into the complex interplay between genetic factors and SUD-related phenotypes in individuals with ADHD. The bidirectional MR analyses suggest a causal relationship between the genetic predisposition to ADHD and increased risk for smoking initiation, emphasizing a potential shared genetic basis.

In the broader context of amphetamine addiction in people with ADHD, the study indirectly contributes by pointing towards the shared genetic architecture between ADHD and certain SUD-related phenotypes. While the study did not specifically address amphetamine addiction, the findings highlight the importance of understanding the genetic factors influencing substance use behaviors in individuals with ADHD.

Taken altogether there is much behavioral and genetic data on amphetamine use, addiction, and ADHD, but very little combining all three. We defined many methods used to tease apart these areas of study and summarized current research on these topics. Mice models have been able to characterize the locomotive and stereotypy behavioral effects of amphetamine usage, indicating involvement of reward circuitry and the associated mechanism involving pathways of transcription factors (Nestler, 2000). Mice models also indicated potential loci of interest in genetic studies, involving hyperthermic response and stereotyped and locomotive behavior; however this still needs further investigation to identify any significant or causal regions (Phillips et al., 2008). Further, genetic correlation studies found significant evidence of correlation between genetic factors associated with ADHD and lifetime usage of drugs (Vilar-Ribó et. al., 2020). Because of the rising rates of amphetamine use, their potential for abuse and the negative outcomes associated therewith, and the greater risk ADHD individuals have for substance abuse, future research in these areas are a priority for public health. Areas of future study should include identifying specific causal genetic regions implicated in amphetamine behavior in mice, researching treatment for amphetamine addiction, and studies specifically investigating the effects and outcomes of amphetamine addiction in ADHD populations (which currently is understudied).

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